

EE3621: Digital Signal Processing

Lecture 21: FIR Filter Specifications & 4 Linear-Phase Types (III B.Tech EEE)

September 9, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** linear phase conditions for symmetric and antisymmetric FIR impulse responses.
 2. **Classify** FIR filters into Types I, II, III, and IV based on length N and symmetry.
 3. **Determine** boundary zero constraints at $\omega = 0$ and $\omega = \pi$ for each FIR type.
 4. **Select** the appropriate FIR type for specific filter applications (LPF, HPF, BPF, BSF, Differentiators, Hilbert Transformers).
 5. **Analyze** zero locations in the complex z -plane.
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2. MATHEMATICAL FOUNDATIONS

2.1 The 4 Types of Linear-Phase FIR Filters

— Type — Symmetry — Length N (Order $M = N - 1$) — Amplitude Function $A(\omega)$ —
Mandatory Boundary Zeros — Suitable Applications — — :—: — :—: — :—: — :—: — :—:
— :—: — — **Type I** — Symmetric ($h[n] = h[N - 1 - n]$) — N Odd (M Even) — $\sum a[k] \cos(\omega k)$
— None — LPF, HPF, BPF, BSF (Universal) — — **Type II** — Symmetric ($h[n] = h[N - 1 - n]$)
— N Even (M Odd) — $\sum b[k] \cos(\omega(k - 1/2))$ — Zero at $\omega = \pi$ ($z = -1$) — LPF, BPF only —
— **Type III** — Antisymmetric ($h[n] = -h[N - 1 - n]$) — N Odd (M Even) — $\sum c[k] \sin(\omega k)$
— Zeros at $\omega = 0$ ($z = 1$) & $\omega = \pi$ ($z = -1$) — BPF, Differentiator, Hilbert — — **Type IV**
— Antisymmetric ($h[n] = -h[N - 1 - n]$) — N Even (M Odd) — $\sum d[k] \sin(\omega(k - 1/2))$ —
Zero at $\omega = 0$ ($z = 1$) — HPF, BPF, Differentiator, Hilbert —

2.2 Proof of Mandatory Zeros

- **Type II at $\omega = \pi$ ($z = -1$):**

$$H(-1) = \sum_{n=0}^{N-1} h[n](-1)^{-n} = \sum_{n=0}^{N/2-1} h[n][(-1)^{-n} + (-1)^{-(N-1-n)}]$$

Since N is even, $N - 1$ is odd $\implies (-1)^{-(N-1-n)} = -(-1)^{-n} \implies H(-1) = 0$. Therefore, a Type II filter always has a zero at $\omega = \pi$ and can **never** be used as a Highpass or Bandstop filter.

2.3 Zero Quadruplet Locations

Because $h[n]$ is real and symmetric/antisymmetric:

$$H(z) = \pm z^{-(N-1)} H(z^{-1})$$

If $z_0 = re^{j\theta}$ is a zero, then:

1. $z_0^* = re^{-j\theta}$ is a zero (conjugate symmetry for real coefficients).
2. $1/z_0 = \frac{1}{r}e^{-j\theta}$ is a zero (linear phase symmetry).
3. $1/z_0^* = \frac{1}{r}e^{j\theta}$ is a zero.

Zeros off the unit circle always occur in groups of 4 (quadruplets).

Pictorial Representations

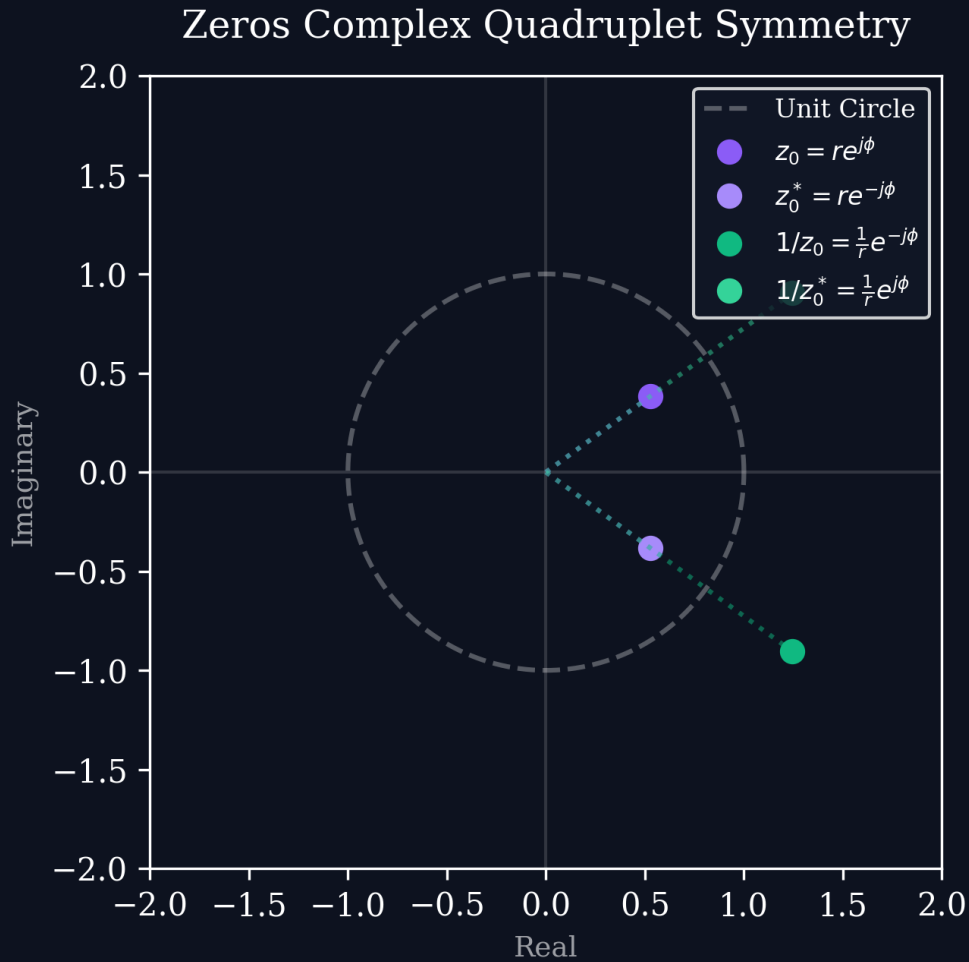


Figure 1: Zero Quadruplet

Frequency Constraints of Linear-Phase FIR Types

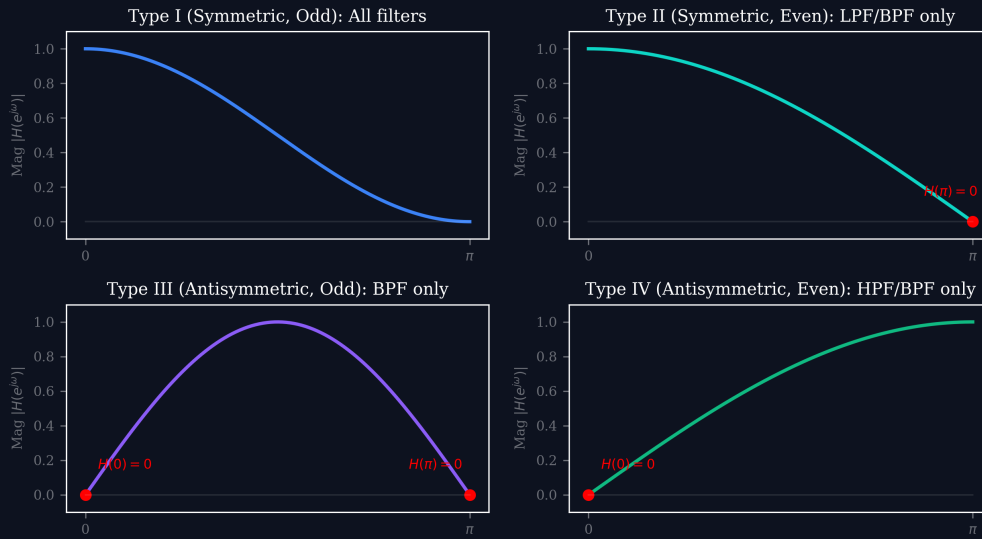


Figure 2: Fir Types

3. WORKED NUMERICAL EXAMPLES

Example 21.1: FIR Type Identification & Zero Locations

Problem: An FIR filter has transfer function $H(z) = 1 + 2z^{-1} + 3z^{-2} + 2z^{-3} + z^{-4}$. (a) Determine its FIR Type, filter length N , and group delay τ_g . (b) Can this filter be used as a Highpass filter?

Solution: (a) Coefficients: $h = [1, 2, 3, 2, 1]$.

- Length $N = 5$ (Odd), Order $M = 4$ (Even).
- Symmetry: $h[0] = h[4] = 1$, $h[1] = h[3] = 2$, $h[2] = 3 \implies$ **Symmetric**.
- Therefore, it is a **Type I FIR Filter**.
- Group delay: $\tau_g = \frac{N-1}{2} = \frac{5-1}{2} = 2$ samples.

(b) Highpass Feasibility: Check $H(e^{j\pi}) = H(-1) = 1 - 2 + 3 - 2 + 1 = 1 \neq 0$. Since Type I has no mandatory zeros at $\omega = \pi$, it can theoretically operate as a Highpass filter (though this specific impulse response has $H(e^{j0}) = 9$ and $H(e^{j\pi}) = 1$, making it a Lowpass filter).

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) State the four types of linear-phase FIR filters and derive the mandatory boundary zeros for Type II, Type III, and Type IV filters. *(9 Marks)* (b) Prove that the roots of a linear-phase FIR filter occur in conjugate reciprocal quadruplets. *(6 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**

* Table of 4 Types with symmetry and length definitions *(3 Marks)* * Mathematical proofs showing $H(e^{j\pi}) = 0$ for Type II, $H(1) = H(-1) = 0$ for Type III, and $H(1) = 0$ for Type IV *(6 Marks)*

- **Part (b):**

* Step-by-step proof using $H(z) = \pm z^{-(N-1)}H(z^{-1})$ and complex conjugation $H^*(z) = H(z^*)$ *(6 Marks)*

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

h = np.array([1, 2, 3, 2, 1])
w = np.linspace(0, np.pi, 500)
H = np.polyval(h[::-1], np.exp(-1j * w))

\title{\textbf{EE3621: Digital Signal Processing} \\ \Large Phase should be exactly -2*w \\
phase = np.unwrap(np.angle(H))
group_delay = -np.diff(phase) / np.diff(w)
print(f"Mean Group Delay: {np.mean(group_delay):.4f} samples (Theoretical = 2.0)")
```